William Rochira

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Profile

Computational scientist with unique cross-disciplinary expertise spanning computational and molecular science. Skilled in both software and hardware development with a systems-level thinking approach. Demonstrated track record of independent research, rapidly mastering complex technical domains to tackle challenging problems. Proven ability to learn and apply cutting-edge computational methods in research environments.

Education

Doctor of Philosophy (PhD) — 2021 – 2025

Computational Biochemistry

University of Oxford, UK

Master of Science (MSc) — 2019 – 2020

Computational Chemistry

University of York, UK

Bachelor of Science (BSc) — 2016 – 2019

Biochemistry — GPA 3.8 / 4.0

University of York, UK

Research Experience

Doctor of Philosophy (PhD)

Supervisor: Professor Philip Biggin

Thesis — 2022 – 2025

Accelerating simulations with analogous hardware: a scalable dataflow architecture for high-performance molecular dynamics

Designed and engineered novel analog-digital hybrid computing systems to accelerate molecular dynamics (MD) simulations. Created *de novo* system architectures and designed custom hardware combining analog circuits, microcontrollers, FPGAs, and mixed-signal ASIC design. Built complete software pipeline, including custom molecular dynamics engine, Python analysis frameworks, and optimisation algorithms incorporating machine learning techniques.

Development of quantum computing algorithms to explore cyclic peptides — 2021

Devised custom implementations of quantum optimisation algorithms with therapeutic potential. Applied variational quantum eigensolvers and quantum approximate optimisation algorithms to explore conformational landscapes of cyclic peptides for drug discovery applications.

Master of Science (MSc)

Supervisors: Professor Kevin Cowtan and Dr. Jon Agirre

Thesis — 2020

Computational validation of macromolecular structural models

Designed software for validating protein models with an intuitive graphical interface. Integrated the tool into CCP4i2, a widely-used software suite used for structural biology research worldwide.

Bachelor of Science (BSc)

Supervisor: Dr. Christoph Baumann

Thesis — 2019

Coarse-grained Monte Carlo simulations to probe confined protein diffusion in the Gram-negative bacterial outer membrane

Developed and optimised coarse-grained Monte Carlo simulations to study confined protein diffusion in bacterial outer membranes, adapting the framework for high-performance cluster computing. Implemented systematic parameter calibration, successfully reproducing experimental single-molecule tracking data.

Computational analysis of mechanical protein unfolding using atomic force microscopy data — 2019

Established a pipeline to automate the analysis of large, high-noise atomic force microscopy datasets, applying machine learning algorithms to detect critical protein unfolding events. Promoted advancements in uncovering disease mechanisms related to protein misfolding and engineering proteins with unique mechanical properties.

Publications

Unconventional computing for the acceleration of molecular simulations — In preparation

Rochira, W. and Biggin, P.C.

Novel methods for the acceleration of protein simulation — 2023

Rochira, W. and Biggin, P.C.

Biophysical Journal

DOI: 10.1016/j.bpj.2022.11.1598

The CCP4 suite: integrative software for macromolecular crystallography — 2022

Agirre, J et al.

Acta crystallographica. Section D, Structural biology

DOI: 10.1107/s2059798323003595

Iris: interactive all-in-one graphical validation of 3D protein model iterations — 2020

Rochira, W. and Agirre, J.

Protein Science

DOI: 10.1002/pro.3955

Intellectual Property

Novel Analog-Digital Hybrid Computing Architecture for Enhanced Molecular Dynamics Simulations — In preparation

Rochira, W. and Biggin, P.C.

Hybrid computing system that integrates analog and digital components to accelerate molecular dynamics simulations. Features custom hardware modules for real-time force calculations and continuous integration, achieving orders-of-magnitude performance improvements over conventional approaches.

Grants and Awards

Medical Sciences Graduate School Studentship — 2021

Funded by EPSRC and IBM

Prestigious and highly competitive studentship awarded annually to a select number of outstanding applicants who demonstrate exceptional academic merit and research potential.

Teaching and Mentoring

High-Performance Computing (HPC) and Simulation Tutor — 2020 – 2021

University of York, UK

Provided one-on-one and group tutoring to undergraduate students, teaching them how to run and analyse biological simulations on HPC systems. Developed materials and tailored sessions to meet the technical needs of each student.

Second- and Third-Year Contact (STYC) — 2017 – 2019

University of York, UK

Assisted first-year students with their transition to university life by providing guidance and mentorship. Responsibilities included providing peer support and offering advice on academic and extracurricular activities.

Seminars

Structural Bioinformatics and Computational Biochemistry (SBCB) Seminar — 2022 – 2024

Chair

Managed logistics, including booking guest speakers and facilitating their visits with introductions, meetings, and tours. Moderated Q&A sessions, prompting insightful discussions between speakers and the audience.

Structural Biology and Molecular Biophysics (SBMB) Seminar — 2021 – 2024

Regular Speaker

Conferences

Oxford Biochemistry Graduate Symposium — 2025

University of Oxford, UK

Poster: *Novel methods for the acceleration of protein simulation*

Oxford-IBM Alliance Symposium — 2023

University of Oxford, UK

Talk: *Unconventional computing for molecular dynamics simulations*

MD in Pharma — 2023

SOAS, London, UK

Attended

Biophysical Society Annual Meeting — 2023

San Diego Convention Center, CA, USA

Poster: *Novel methods for the acceleration of protein simulation*

UK-QSAR Autumn Meeting — 2022

Francis Crick Institute, London, UK

Attended

Oxford-IBM Alliance Symposium — 2022

University of Oxford, UK

Talk: *Analog computing for molecular dynamics simulations*

CCP4 Study Weekend — 2021

East Midlands Conference Centre, UK

Talk: *Iris: interactive all-in-one graphical validation of 3D protein model iterations*

CCP4 Study Weekend — 2020

East Midlands Conference Centre, UK
Attended

CFS/ME Research Collaborative Conference — 2018

Future Inn Bristol, UK
Attended

Professional Experience

Acturis

Technical Business Analyst — 2020

Designed and implemented large-scale machine learning models for insurance customer churn prediction, developing real-time risk scoring systems deployed in commercial production across major UK insurance databases, processing industry-wide policy data.

TrainTaxi

Web Developer and Server Administrator — 2019 – 2020

Built full-stack web application with MySQL database, PHP backend, and HTML5 frontend control panel. Maintained and re-engineered existing Django and Ruby on Rails systems.

Consulting Software Developer — 2014 – 2019

Developed automation pipelines in Python and native macOS applications in Swift, comprising an extensive library of data analysis tools.

Selected Technical Projects

Machine Learning for Molecular Dynamics Acceleration — 2022 – 2024

Physics-Informed Neural Networks

Developed integer-only neural networks optimised for FPGA deployment, achieving MHz-frequency force field evaluation through hardware-software co-design with quantised architectures for real-time molecular dynamics acceleration.

Deep Reinforcement Learning

Designed multi-agent reinforcement learning system for dynamic hardware resource allocation in molecular simulations, implementing intelligent module assignment strategies that balance simulation accuracy with hardware utilisation efficiency.

Transformer Networks

Built attention-based models for real-time molecular energy prediction from molecular dynamics trajectories, with optimised architecture achieving high-speed inference suitable for hardware-accelerated systems.

Poker Bot — 2018 – 2020

Engineered sophisticated poker-playing system implementing counterfactual regret minimisation for optimal strategy computation, and graph neural networks for multi-player game state modelling. Developed generative adversarial networks for opponent simulation and Monte Carlo engines for real-time probability calculations. Used direct memory reading for game-state extraction to achieve millisecond-latency.

Algorithmic Trading System

2017 – 2019

Developed multi-exchange cryptocurrency trading system using long short-term memory networks for price prediction. Implemented reinforcement learning agents with Monte Carlo tree search for strategy optimisation. Built real-time processing system handling WebSocket feeds with millisecond-latency decision-making. Developed comprehensive backtesting framework processing 18+ months of tick-level data, achieving several successful live trading deployments with automated risk management.

Skills

Programming

Highly proficient:	Python
Experienced:	C, Verilog, PHP, JS/Node.js, SQL, Bash
Familiar:	C++, C#, Java

Data Science

Python frameworks:	NumPy, SciPy, Pandas, Matplotlib, TensorFlow, Keras, scikit-learn
Databases:	SQL (MySQL, Oracle SQL)

Machine Learning

Neural networks:	Physics-informed networks, transformers, convolutional networks
Reinforcement learning:	Multi-agent systems, Monte Carlo methods, policy optimisation
Computer vision:	OpenCV, pattern recognition, real-time image processing
Hardware deployment:	Neural network quantisation, FPGA optimisation, real-time inference

Electronic Engineering

Analog circuit design:	SPICE, LTspice
Digital circuit design:	Logisim
Hybrid circuit design:	Proteus, TINA
PCB design:	KiCad
Embedded systems:	Analog/digital hybrid systems Microcontroller programming (C, MicroPython)
FPGA development:	HDL (Verilog / SystemVerilog) Simulation (Verilator) Synthesis and implementation (Vivado)
ASIC development:	ASIC design and implementation (Cadence Virtuoso)

Other

HPC:	Slurm and Sun Grid Engine (now Oracle)
CAD:	Blender
Version control:	Git
Adobe suite:	Photoshop, Illustrator
Molecular simulations:	Gromacs, MDAnalysis, BioPython
Molecular modelling:	CCP4, Phenix, PyMOL, ChimeraX

Languages

English, Italian